

**10. Implement transformation-based tagging using a set of transformation rules, apply a simple rule to tag words using python.**

```

sentence = "The dog is running quickly"

words = sentence.split()

# Initial tagging
tags = []

for word in words:
    tags.append((word, "NN"))

# Apply transformation rules
new_tags = []

for word, tag in tags:
    if word.endswith("ing"):
        tag = "VBG"
    elif word.endswith("ly"):
        tag = "RB"
    elif word.lower() in ["is", "am", "are", "was", "were"]:
        tag = "VB"
    elif word.lower() in ["the", "a", "an"]:
        tag = "DT"
    new_tags.append((word, tag))

print("Transformation-Based POS Tags:")
print(new_tags)

... Transformation-Based POS Tags:
[('The', 'DT'), ('dog', 'NN'), ('is', 'VB'), ('running', 'VBG'), ('quickly', 'RB')]

```